

for every electronic device. In case of instruments used for field applications, it is not practical to use bulky cooling systems as this increases the size and weight, and reduces the mobility of the testing instrument. Since such devices operate for long hours in rugged environmental conditions, these are placed inside a chassis that has an electrically-insulating and thermally-conducting layer of coating. This helps in the safe dissipation of large amounts of heat generated in an RF test and measurement instrument.

MIMO. Multiple-input and multiple-output (MIMO) is the method of using multiple antennae at both the transmitter and receiver sections in radio communication technology. It increases the link capacity, link reliability and spectral efficiency by exploiting multipath propagation, which was previously considered as interference. The fact that MIMO needs no extra bandwidth or transmit power in order to increase the data throughput and link range made most of the modern researches to be concentrated on this area.

The physical layer performance characterisation can be performed by the MIMO PXI test solution from Keysight Technologies. This solution provides flexibility for a validation engineer to try out a variety of test cases and ensure that the WLAN design works perfectly. In case of MIMO

Instruments you need

Let us take a look at the basic test and measurement equipment required to evaluate an RF device.

RF power meters. Spectrum analysers generally have the capability of measuring up to about 100mW of power. For measuring values above that, we generally use a power meter. Power meters measure the transmission power of an RF device over a wide frequency range and display high-accuracy results. But a power meter has a limited amplitude measurement range and gives no information on the frequency composition of the signal.

RF signal/spectrum analyser. This device measures the magnitude of an input signal in a frequency domain using narrowband detection techniques. Though the accuracy level is low compared to power meters, the spectrum/signal analysers can measure values as low as -150dBm. Spectrum analyser checks if the transmitter is generating appropriate power spectrum and the vector signal analyser measures how much modulation distortion is created by the device.

Network analyser. It measures the response of a device at radio frequencies and displays that in rectangular co-ordinates, polar co-ordinates or a Smith chart. It lets you determine parameters such as transmission losses and gains, return loss, impedance match, phase measurements and group delay over a wide range of frequencies for components like filters and amplifiers. By evaluating the characteristics, one can determine the behaviour of the component in the intended RF device.

RF signal generators. These instruments are used to create a stimulus signal for testing components, devices or systems. It either generates the signals internally or uses the signal created externally and downloaded into the instrument. These are used for receiver sensitivity tests, bit error rate (BER) tests, adjacent channel rejection and two-tone inter-modulation tests.



Keysight M937xA network analyser

The new low-cost switching and instrumentation architecture

In September 2014, the AdvancedTCA Extensions for Instrumentation and Test (AXIe) Consortium announced a new standard named AXIe-0, which is expected to broaden the use of AXIe, targeted for low-cost modular instrumentation and switching applications. It is an ideal platform for users who require major capabilities of AXIe-1 at an affordable price.

AXIe-0 is completely upward compatible with the existing specification AXIe-1, thereby making it a subset of the mainstream standard. This means that the AXIe-0 module can be used in AXIe-1 chassis. Features like module size (3.05×27.94×32.26 cms), board area, slot scalability and horizontal and vertical configurations of AXIe-0 are the same as the AXIe-1 base architecture.

The cost has been cut down by the use of LAN interface, single power supply voltage of -48V, reducing zone 2 connectors from five to two, and elimination of zone three connectors and intelligent platform management interface (IPMI).

AXIe-0 can be used for applications that require AXIe's space, power and form factor, but do not necessarily need a high-speed data infrastructure. Some of the application areas of the new specification include large switching systems and RF interface units for military and aerospace electronic functional test, VME xExtensions for Instrumentation (VXI) replacement in mil/aero, custom instrumentation and functionalities by system integrators or users, general purpose (loads, frequency counters, etc) and data acquisition.

beamforming, even a slight inaccuracy in phase relationship will lead to irrelevant and misleading results. MB5 MIMO beamforming testing system was introduced by Spirent to automate the phase calibration in order to meet the tight tolerance values required by MIMO beamforming.

Femtocell. Femtocell is a small, low-power wireless access point that is used to provide better coverage for cellular networks inside homes, office buildings and public spaces. It is one of the recent advancements in mobile communication; the device resembles a wireless router.

E6650A EXM wireless test set by Keysight Technologies and EAST